

Design & Implementation of a Segmented Enterprise Network

Inter-VLAN Routing & VLSM IP Subnetting Case Study

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1. Project Overview

The objective of this project is to design, configure, and verify a segmented enterprise network infrastructure utilizing Cisco Packet Tracer. The network is logically partitioned into three functional departments (Engineering, HR, and Sales) at Layer 2 and Layer 3. This implementation minimizes broadcast domains, optimizes localized performance, and enhances baseline security by isolating departmental traffic while enabling secure, deterministic Inter-VLAN routing across specific physical interfaces.

2. Network Topology

The infrastructure uses a traditional two-tier collapsed core architecture design consisting of a central router (Router R1) connected to a main switch (Switch SW1). Separate physical interfaces on the router are assigned to map each individual VLAN directly, establishing a highly stable environment for inter-departmental communications.

- **VLAN 10 (Engineering):** Designated hosts belonging to the engineering division.
- **VLAN 20 (HR):** Designated hosts belonging to the human resources division.
- **VLAN 30 (Sales):** Designated hosts belonging to the sales division.



[IMAGE PLACEHOLDER: Full Network Topology Layout showing all links green/up]

3. IP Subnetting & Addressing Scheme

To ensure strict host optimization and zero subnet overlapping, Classless Inter-Domain Routing (CIDR) is applied with a fixed /26 network mask (providing up to 62 usable host addresses per block). The addressing parameters are documented below:

VLAN ID	Department Name	Subnet / CIDR	Network ID	Gateway IP (Router Interf.)	Usable Host IP Range	Broadcast Address
10	Engineering	10.0.0.0/26	10.0.0.0	10.0.0.62	10.0.0.1 - 10.0.0.61	10.0.0.63
20	HR	10.0.0.64/26	10.0.0.64	10.0.0.126	10.0.0.65 - 10.0.0.125	10.0.0.127
30	Sales	10.0.0.128/26	10.0.0.128	10.0.0.190	10.0.0.129 - 10.0.0.189	10.0.0.191

4. Technical Configuration Script

4.1 Switch VLAN Taggings & Port Range Assignments

Ports are mapped efficiently utilizing the Cisco IOS interface `range` parameter, accelerating delivery and switching ports immediately into `access` mode. If the target VLAN does not explicitly exist, the IOS automatically initializes it:

```
SW1> enable
SW1# configure terminal

! Allocate Engineering interfaces and enforce access configuration
SW1(config)# interface range g0/1, f3/1, f4/1
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config-if-range)# exit

! Allocate HR interfaces and enforce access configuration
SW1(config)# interface range g1/1, f5/1, f6/1
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# exit

! Allocate Sales interfaces and enforce access configuration
SW1(config)# interface range g2/1, f7/1, f8/1
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 30
SW1(config-if-range)# end
```

 [IMAGE PLACEHOLDER: Switch CLI Screenshot showing "do sh vl br" execution]

4.2 Router Interface Configurations

The interface metrics are activated by applying the Last Usable Host IP to work seamlessly as the default routing point for all outgoing data:

```
R1# configure terminal
R1(config)# interface gigabitethernet0/0
R1(config-if)# ip address 10.0.0.62 255.255.255.192
R1(config-if)# no shutdown

R1(config)# interface gigabitethernet0/1
R1(config-if)# ip address 10.0.0.126 255.255.255.192
R1(config-if)# no shutdown

R1(config)# interface gigabitethernet0/2
R1(config-if)# ip address 10.0.0.190 255.255.255.192
R1(config-if)# no shutdown
```

Engineering Troubleshooting & Field Notes

- 1. Subnet Broadcast Resolution (Bad Mask Error):** During initial interface assignment, the router rejected inputs with a Bad mask /26 for address error message. Upon evaluation of the binary block size boundaries, it was determined that the targeted addresses (.63, .127, and .191) matched the subnets' Broadcast Addresses. The configurations were corrected immediately using the Last Usable Host IP address (.62, .126, and .190), bringing up the interfaces successfully.
- 2. CLI Syntax Correction:** Captured minor typo indicators (e.g., switchportt, no shutt) marked by the IOS ^ syntax flag. Typos were systematically resolved in real-time, executing valid commands to guarantee robust running configurations.

5. Verification & Connectivity Testing

To ensure absolute infrastructure readiness, multiple verification methods were deployed:

- Running `show ip interface brief` to verify that all dedicated physical Layer 3 router slots indicate an **up/up** operational status.
- Running `show vlan brief` to guarantee proper port bindings on Switch SW1.
- Executing cross-network Ping requests to validate End-to-End Inter-VLAN Routing data plane deliveries.

 [IMAGE PLACEHOLDER: PC Command Prompt Screenshot indicating successful Ping verification]

How to Insert Your Actual Screenshots into the Final Document

Since a compiled PDF is a final visual format, you cannot directly drop raw image files into a pre-compiled PDF. To inject your actual Cisco Packet Tracer screenshots, follow these simple steps:

1. Open this document in any modern PDF Editor (e.g., Adobe Acrobat Pro, Foxit PhantomPDF, or free online tools like **Sejda PDF Editor** or **PDFescape**).
2. Locate the dashed [IMAGE PLACEHOLDER] boxes provided above.
3. Select the "**Add Image**" tool from your editor's menu, click inside the box, and upload your corresponding screenshot.
4. Save the document. Your professional portfolio case study is now fully localized and complete!

--- End of Technical Project Case Study ---